

The Half-Axis Problem: On the Asymmetric Interpretation of a Symmetric Temporal Formalism

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Abstract

Physics defines time as a real-valued variable, $t \in \mathbb{R}$, and exploits the full real line in its computational apparatus: two-sided Fourier transforms, time-ordered correlation functions, the time-reversal operator, Feynman propagators, Green's functions, analytic continuation. Yet in its interpretive framework, physics treats the physically meaningful domain of t as the half-line $[0, +\infty)$ or, locally, as the forward direction alone. This paper isolates this discrepancy as a distinct foundational problem—the half-axis problem—and argues that it is not a variant of the arrow-of-time problem. The arrow problem asks why processes are asymmetric within an accepted temporal domain; the half-axis problem asks what licenses the restriction of the interpreted domain itself. I examine the three standard justifications for the restriction—thermodynamic, causal, and cosmological—and show that each establishes at most a physical asymmetry within the domain, not a restriction of the domain, and that each admits of a counter-objection internal to its own terms that has not been met. I then develop an inductive argument from the history of physical quantities: temperature, energy, spatial position, negative frequency, the negative-energy solutions of the Dirac equation, and negative curvature were each initially treated as positive-definite or meaningless on the basis of regime-limited intuition, and in each case extension to the full mathematical range revealed new physics. The paper concludes that the half-axis restriction is an unjustified convention, and that the point $t = 0$ —the unique fixed point of the time-reversal operator, the additive identity of the temporal group, and the boundary of the interpreted domain—possesses a structural status that the interpretive framework of physics has not acknowledged. Whatever the correct interpretation of the complete axis turns out to be, the current asymmetric interpretation is not entitled to its own self-evidence.

Keywords: time asymmetry, temporal domain, half-axis problem, arrow of time, time-reversal symmetry, negative time, formal domain, interpretive domain, $t = 0$ fixed point, regime-limited intuition

1. Introduction: A Discrepancy Hidden in Plain Sight

In the dominant continuous-time formulations of physics developed since 1687, the temporal variable t is treated as a real number, defined on the complete real line $(-\infty, +\infty)$, and the formal machinery of the discipline—differentiation, integration, Fourier analysis, the symmetry operations of field theory—makes essential use of the completeness of that domain. In every interpretive statement physics makes about time, the negative half of that domain is treated as physically empty: not as a region where physics fails to apply, but as a region that is not so much as mentioned.

This paper argues that the discrepancy deserves a name and an analysis. I will call it the half-axis problem: the problem of what, if anything, licenses the restriction of the physically interpreted temporal domain to half of the mathematically employed one.

The half-axis problem must be distinguished at the outset from the arrow-of-time problem, with which it is easily conflated. The arrow problem—the subject of an enormous literature from Boltzmann (1877) through Reichenbach (1956), Price (1996), Albert (2000), and Callender (2021)—asks why physical processes exhibit temporal asymmetry given that the underlying dynamical laws are (CPT considerations aside) time-reversal invariant (Sklar, 1993; Earman, 1974). The arrow problem takes the temporal domain as given and asks about the behaviour of systems within it. The half-axis problem is prior: it asks about the constitution of the interpreted domain itself. One can hold any position whatever on the arrow problem—Boltzmannian statistical, interventionist, Past-Hypothesis-based, perspectival—and the half-axis problem remains untouched, because every one of those positions is a thesis about asymmetric behaviour on an axis whose interpretive truncation none of them addresses. A direction is a structure on a domain already given; the half-axis problem concerns the domain, not the direction, and is therefore prior to every version of the arrow debate.

Two terms are used throughout this paper with a precision their ordinary usage does not always carry. The formal domain of a physical quantity is the set of values over which the quantity's mathematical definition ranges—the domain specified by the equations, the integration variables, the symmetry operators, and the analytic instruments of the theory. The interpretive domain is the set of values that the discipline's explanatory and representational practice treats as physically meaningful—the values that appear in physical narratives, that are assigned physical referents, and whose limits are treated as limits of the phenomenon rather than limits of the model. The half-axis problem is the problem that arises when these two domains diverge: when the formal domain is $(-\infty, +\infty)$ and the interpretive domain is $[0, +\infty)$, and when no argument has been given for the divergence.

The structure of the paper is as follows. Section 2 documents the extent of physics's formal commitment to the complete axis across six distinct formalisms—classical dynamics, the Fourier apparatus, the

Green's function and propagator framework, the time-reversal operator, Minkowski spacetime, and the path integral. The point is elementary in each individual case, but the cumulative force of the survey is underappreciated: the complete axis is not an optional extension of physics's mathematical apparatus but its native and ineliminable domain. Section 3 examines the three standard justifications for the interpretive restriction—thermodynamic, causal, and cosmological—and argues that each commits the same category error: treating a defeasible physical asymmetry within the domain as though it effected an infeasible mathematical restriction of the domain. Each justification is examined on its own terms, and each is shown to admit a counter-objection internal to those terms that the justification has not met. Section 4 develops the inductive argument from the history of quantity extensions: temperature, energy, spatial position, negative frequency, Dirac's negative-energy solutions, and negative curvature each followed a five-stage pattern in which a regime-limited truncation was eventually recognised as such and the extension to the full range revealed new physics. The pattern is distilled into an explicit epistemic heuristic whose application to the temporal axis shifts the burden of argument. Section 5 analyses the structural status of $t = 0$ under the complete-axis reading—its topological, algebraic, and dynamical characterisations—and examines the asymmetry between the conventional treatment of spatial and temporal origins. Section 6 distinguishes the present argument from Price's Archimedean programme, states with precision what has and has not been established, and identifies three directions for further investigation.

2. The Formal Commitment to the Complete Axis

The claim of this section is descriptive: mainstream physics is formally committed, pervasively and ineliminably, to the domain $t \in (-\infty, +\infty)$. The argument proceeds in four layers of increasing structural depth—from the equations of motion, to the analytic instruments that solve them, to the propagators that encode physical content, to the symmetry operations that define the laws themselves—because the completeness of the temporal axis is not a feature of one formalism but a presupposition of all of them. Two further formalisms, Minkowski spacetime and the path integral, are then examined separately because each makes the formal commitment in a geometrically and conceptually distinct way. The intention throughout is to establish that the complete axis is not an optional extension of the formalism but its native domain, and that the interpretive restriction to the half-axis enters, without argument, only at the level of interpretation.

2.1 Classical and Quantum Dynamics

Newtonian mechanics defines dynamics through derivatives d/dt whose domain is unrestricted (Newton, 1687). The second law, $F = ma = m \, d^2x/dt^2$, is an equation on $\mathbb{R}$; any restriction to $t \geq t_0$ is an initial condition imposed on the solutions, a fact about the problem being solved, not about the equations

themselves. This distinction—between the domain of an equation and the domain of a particular solution—is elementary but consequential. An initial condition at $t = t_0$ selects a solution from the family of all solutions; it does not modify the equation, and the equation retains its full temporal domain regardless of which initial condition is imposed. The same holds for Hamilton's equations, $dp/dt = -\partial H/\partial q$ and $dq/dt = \partial H/\partial p$, whose Hamiltonian flow is defined for all $t \in \mathbb{R}$; for the Euler–Lagrange equations, $d/dt(\partial L/\partial \dot{q}) - \partial L/\partial q = 0$, which are equally unrestricted; and for every other formulation of classical dynamics, including the Hamilton–Jacobi equation and the Liouville equation for phase-space distributions. The formal commitment established here—that the temporal variable of the equations of motion ranges over $\mathbb{R}$ —is not a peculiarity of classical mechanics but carries through without modification into the field equations that supersede it.

Maxwell's equations, $\partial_\mu F^{\mu\nu} = J^\nu$ and $\partial_\alpha [\epsilon_{\alpha\beta\gamma} F^{\beta\gamma}] = 0$, are partial differential equations in a temporal variable running over $\mathbb{R}$ (Maxwell, 1865). The restriction to $t > t_0$ that appears in the formulation of initial-value problems for electromagnetic fields is an initial condition imposed by the physics of the specific problem—specifying the field configuration at some reference time—not a constraint on the temporal variable appearing in the equations. The equations themselves remain defined for $t < t_0$, and their retarded and advanced solutions, which propagate influence forward and backward in time respectively, both exist as mathematical solutions on the complete temporal axis. The same structure persists in the passage to quantum mechanics, where the temporal variable of the field equations becomes the temporal variable of the state-evolution equation, with the same unrestricted domain.

The Schrödinger equation, $i\hbar \partial\psi/\partial t = \hat{H}\psi$, evolves the state $\psi(x, t)$ for all real t (Schrödinger, 1926). The unitary time-evolution operator $U(t, t_0) = \exp(-i\hat{H}(t - t_0)/\hbar)$ is defined for all $t \in \mathbb{R}$, including $t < t_0$, where it implements time-reversal of the evolution. In scattering theory—the empirical core of particle physics—this two-sided character is not merely formal but physically essential. The in-state $|\psi_{\text{in}}\rangle$ is defined by the asymptotic condition $\lim_{t \rightarrow -\infty} \|\psi(t) - |\phi_{\text{in}}(t)\rangle\| = 0$, where $|\phi_{\text{in}}(t)\rangle$ is a free-particle state at $t \rightarrow -\infty$. The S-matrix, $S = \lim_{t \rightarrow +\infty, t_0 \rightarrow -\infty} U_I(t, t_0)$, is explicitly a map from the asymptotic past to the asymptotic future, constructed by integrating the interaction Hamiltonian over the entire real line. The limit $t_0 \rightarrow -\infty$ is not a notational convenience; it is a physical requirement that the incoming particles be far enough separated in the past that their mutual interaction is negligible. The entire scattering formalism—which underlies the calculation of every cross-section, decay rate, and scattering amplitude in the Standard Model—is a two-ended construction on the complete temporal axis. The equations of motion, from classical mechanics through quantum field theory, share a single formal commitment: the temporal variable is defined on $\mathbb{R}$, and initial conditions are conditions on solutions within that domain, not restrictions of the domain itself. The analytic instruments used to extract physical content from these equations make the same commitment, independently and for their own structural reasons.

2.2 The Two-Sided Fourier Transform

The two-sided Fourier transform, $F(\omega) = \int_{-\infty}^{+\infty} f(t) e^{-i\omega t} dt$, is not a convenience of computational practice but a load-bearing structure whose physical content depends ineliminably on the integral running over the complete line. The claim that a temporal signal decomposes into pure harmonic oscillations—the foundational claim of frequency-domain analysis in quantum theory, field theory, optics, acoustics, and signal processing—requires the two-sided integral. The mathematical content of this claim is the Plancherel theorem and its generalisations, which hold for functions defined on $\mathbb{R}$; they do not hold, without substantial modification, for functions defined on a half-line.

The one-sided analogue of the Fourier transform is the Laplace transform, $L\{f\}(s) = \int_0^{+\infty} f(t) e^{-st} dt$, a different mathematical object with different analytic properties, different convergence conditions, and a different physical interpretation. While the Laplace transform has its own physical applications—primarily in the analysis of causal linear systems, precisely because its one-sided character encodes causality by construction—it is not the instrument that physics uses as its default tool in quantum mechanics, quantum field theory, or relativistic wave equations. Physics uses the two-sided Fourier transform, and in doing so asserts, at every use, that its functions are defined and integrable over $(-\infty, +\infty)$.

The interpretive restriction to $t \geq 0$ is in direct tension with this choice of instrument. A function whose domain is intrinsically restricted to $t \geq 0$ does not possess the same two-sided Fourier representation without an additional extension prescription; at best it has a Fourier transform of its extension by zero to $t < 0$, which is a different function with different spectral properties. Physics uses the two-sided transform ubiquitously and correctly—because its functions are genuinely defined on the complete axis—and then, in interpretive statements, treats those same functions as though they were defined only on the half-line. The inconsistency is not in the mathematics, which is self-consistent; it is between the mathematics and the interpretation. That the analytic instruments used to solve the equations of physics make the same formal commitment as the equations themselves is the second layer of the argument.

2.3 Green's Functions and Propagators

The Green's function formalism is the systematic method for solving linear differential equations with sources throughout classical and quantum physics, and it requires the complete temporal axis both in its formulation and in the physical identities it generates. The Green's function $G(t, t')$ for a differential operator $(\partial/\partial t - L)$ satisfies $(\partial/\partial t - L) G(t, t') = \delta(t - t')$, where δ is the Dirac delta, and its integral representation is obtained by Fourier transform over the complete real frequency axis: $G(\omega) = 1/(-i\omega - L(\omega))$. The standard frequency-space representation—in which the poles of $G(\omega)$ in the complex frequency plane encode the natural frequencies and decay rates of the system—depends on the two-sided Fourier construction running over all of $\mathbb{R}$.

The retarded and advanced Green's functions, G_R and G_A , are the two physically relevant special cases. The retarded Green's function, $G_R(t, t') = -\theta(t - t') [A(t), A(t')]/(i\hbar)$, where θ is the Heaviside step function, vanishes for $t < t'$ and is used to propagate solutions forward in time from a source at t' . The advanced Green's function, $G_A(t, t') = \theta(t' - t) [A(t), A(t')]/(i\hbar)$, vanishes for $t > t'$ and propagates solutions backward in time. Both are defined on the complete real line; the Heaviside step functions encode the directionality of propagation as a condition within the domain, not as a restriction of the domain. That G_R vanishes for $t < t'$ does not mean that $t < t'$ is outside the domain of G_R ; it means that G_R takes the value zero there, which is a fact about the function, not about its domain.

The relation between G_R and G_A is an identity on the complete frequency axis: $G_R(\omega) - G_A(\omega) = -2\pi i A(\omega)$, where $A(\omega)$ is the spectral function. This identity is the basis of the fluctuation-dissipation theorem, which connects the response of a system to external perturbations with its equilibrium fluctuations, and of the Kramers–Kronig relations, which connect the real and imaginary parts of the response function through the requirement of analyticity in the upper half of the complex frequency plane. Both of these results—among the most general and widely applied results in many-body physics—are theorems on the complete frequency axis whose derivation requires that the Fourier transform be taken over all of $\mathbb{R}$. They would be unavailable, or would require substantial reformulation, if the temporal axis were genuinely restricted to $t \geq 0$. That the analytic instruments used to construct and solve the propagators share the same formal commitment as the equations of motion is not a coincidence of notation; it reflects the fact that the complete axis is the native domain of both. The same commitment appears, in its most operationally direct form, in the symmetry operations through which physics defines and tests its fundamental laws.

The Feynman propagator, $\Delta_F(x - y) = \langle 0 | T \{ \phi(x) \phi(y) \} | 0 \rangle$, where T denotes the time-ordering operator, is the central object of perturbative quantum field theory. Its poles in the complex momentum plane—at $p^2 = m^2 - i\epsilon$ for the Klein–Gordon field—encode the particle spectrum of the theory, and its perturbative expansion in powers of the coupling constant generates the Feynman diagram expansion whose predictions are compared to experiment. The time-ordering prescription, $T\{A(t)B(t')\} = \theta(t - t') A(t)B(t') + \theta(t' - t) B(t')A(t)$, uses step functions to select either the retarded or advanced ordering depending on the temporal relation between t and t' ; it is defined for all real t and t' , and the combination of the two orderings is what gives the Feynman propagator its distinctive analytic structure—poles with the $i\epsilon$ prescription—which is essential for the calculation of S-matrix elements.

The Wick rotation, $t \mapsto i\tau$ with $\tau \in \mathbb{R}$, passes from the Minkowski propagator to the Euclidean propagator by analytic continuation in the complex time plane. The justification for this procedure—that the Minkowski and Euclidean path integrals compute the same physical amplitudes—relies on the analytic structure of the Minkowski propagator as a function on the complete real temporal axis, and on the fact that the Euclidean time τ runs over all of $\mathbb{R}$ just as the Minkowski time t does. The Wick rotation presupposes the complex-time neighbourhood of the real axis, which is only defined if the real axis

itself is complete. A function defined only on $[0, +\infty)$ does not have a Wick rotation in the standard sense; the analytic continuation would reach into negative Euclidean times, which correspond to positive Minkowski times, without any corresponding structure on the negative Minkowski half-axis. That propagators make the same formal commitment to the complete axis as the equations and the analytic instruments is the third layer of the argument; the final layer concerns the symmetry operations that govern the laws themselves.

2.4 The Time-Reversal Operator and Its Physical Consequences

The time-reversal operator $T: t \mapsto -t$ is the most explicit and irreducible formal commitment to the negative half-axis in the entire apparatus of physics. It is not a device for computing negative-time quantities; it is an operator that maps the temporal variable onto its negative values, and its physical significance is the investigation of whether the laws of physics are symmetric under this mapping. That investigation is one of the precision enterprises of modern physics, and its results are experimentally confirmed to high precision.

In non-relativistic quantum mechanics, the time-reversal operator is an antiunitary operator $\Theta = UK$, where U is a unitary operator and K denotes complex conjugation. Acting on a state $|\psi\rangle$, Θ reverses the sign of momenta and spins—the quantities that are odd under time reversal—while leaving positions unchanged. The Schrödinger equation is invariant under the combined action of $t \mapsto -t$ and Θ : if $\psi(x, t)$ is a solution, so is $\Theta\psi(x, -t)$. This invariance is the expression of T -symmetry in non-relativistic quantum mechanics, and it is the baseline from which T -violation in relativistic theories is measured as a departure.

In quantum field theory, the CPT theorem—the Lüders–Pauli theorem—establishes that any local Lorentz-invariant quantum field theory is invariant under the combined operation CPT, where C is charge conjugation and P is parity inversion, from the axiomatic assumptions of the theory alone (Streater & Wightman, 1964). The CPT theorem does not require T -invariance individually; it requires only the combination. The observed CP violation in neutral kaon decay, established by Christenson, Cronin, Fitch and Turlay (1964), therefore entails compensating T -violation by the CPT theorem: if CP is violated and CPT is exact, then T must be violated by the same amount. Direct experimental observation of T -violation—not merely inferred from CP-violation via CPT—was achieved by the BaBar collaboration (2012) in the B-meson system, at a significance of fourteen standard deviations. The measurement establishes, from the observation of an asymmetry in decay rates under the exchange of initial and final states, that the laws governing B-meson decay are not invariant under time reversal.

Physics does not merely tolerate the negative half-axis; it measures the symmetry properties of the laws under reflection into it, to a precision of parts per thousand, and it identifies physical effects—the matter–antimatter asymmetry of the universe—with the failure of exact T -symmetry. The point for the half-axis problem is this: the time-reversal operator T is defined as the map $t \mapsto -t$, and the experimental

programme that probes T-symmetry presupposes the complete temporal axis as the formal structure within which that map is meaningful. The BaBar collaboration does not measure a quantity defined only on $t \geq 0$; it measures an asymmetry whose definition requires that the reflection $t \mapsto -t$ be a well-formed operation—which it is only if the negative half-axis is a legitimate part of the temporal variable's domain. An interpretive framework that deletes the negative half-axis is therefore in tension not merely with the abstract formalism of quantum field theory but with the formal presuppositions of the experimental programme that uses the time-reversal operator as a precision instrument. From the complete-axis reading, T-violation is a fact about the asymmetry of physics under temporal reflection; from the half-axis reading, the operator that produces that reflection maps the variable outside its interpreted domain, which is an incoherence the half-axis framework has no resources to resolve. That incoherence does not surface in practice because the complete-axis reading is what physics actually uses in calculation—the interpretive restriction is abandoned the moment the mathematics requires the negative half-axis, which is constantly. The gap between what physics computes and what it claims to interpret runs through every use of the time-reversal operator. The next two formalisms show the same gap at the level of geometry and of the quantum theory of the universe as a whole.

2.5 Minkowski Spacetime and the Lorentzian Signature

In special and general relativity, time enters the geometry as one coordinate among four, distinguished from the spatial coordinates not by the extent of its range but by the sign of its contribution to the spacetime interval. In the $(-, +, +, +)$ signature convention, the metric takes the form $ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2$ in Cartesian coordinates, and the temporal coordinate t runs over $\mathbb{R}$ by the same mathematical licence as x , y , and z . The metric signature determines the causal character of displacements—timelike intervals with $ds^2 < 0$, null intervals with $ds^2 = 0$, spacelike intervals with $ds^2 > 0$ —but it does not restrict the range of the temporal coordinate. A spacetime event is a point $(t, x, y, z) \in \mathbb{R}^4$, and the geometry of Minkowski spacetime is the geometry of $\mathbb{R}^4$ equipped with the metric $\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$.

The algebraic structure of Minkowski spacetime makes the formal commitment to the complete temporal axis precise. Minkowski spacetime $(\mathbb{R}^4, \eta_{\mu\nu})$ is a four-dimensional real vector space, closed under addition and scalar multiplication of four-vectors. If t is restricted to $[0, +\infty)$ while x , y , z remain unrestricted, the resulting set $\{(t, x, y, z) : t \geq 0, x, y, z \in \mathbb{R}\}$ is not a vector space: the sum of two events with $t > 0$ remains in the restricted set, but the negation of an event, which is required for the vector space structure, has $t < 0$ and falls outside it. The resulting set is also not a group under addition, since the additive inverse of any event with $t > 0$ is not in the set. It is, in short, not the arena of special relativity. The geometry of Minkowski spacetime is constitutively the geometry of $\mathbb{R}^4$ with the Lorentzian metric; restricting t to the half-line while maintaining the rest of the structure would produce an object that is not Minkowski spacetime.

This observation connects to the treatment of spatial coordinates. No physicist restricts x to $[0, +\infty)$ on the grounds that lengths are positive; the spatial coordinate x is understood to range over $\mathbb{R}$, with its sign indicating direction relative to a chosen origin. The formal situation for t is analogous: the temporal coordinate ranges over $\mathbb{R}$, with its sign functioning as a coordinate label relative to a chosen temporal origin—though its causal role differs from that of spatial coordinates by virtue of the metric signature, a difference that affects causal structure without restricting the coordinate's range. The asymmetry in interpretive practice— x freely ranging over $\mathbb{R}$, t restricted in interpretation to $[0, +\infty)$ —is not grounded in any formal difference between the coordinates. Both are real numbers; both enter the metric on the same footing, differing only in the sign of their contribution; both range over $\mathbb{R}$ in the mathematical definition of the spacetime. The asymmetry is an interpretive habit, and its grounds have not been stated.

2.6 The Path Integral

The Feynman path integral, $\langle x_f, t_f | x_i, t_i \rangle = \int Dx(t) \exp(iS[x]/\hbar)$ (Feynman, 1948), is a sum over all paths $x(t)$ connecting the initial configuration (x_i, t_i) to the final configuration (x_f, t_f) , weighted by the phase factor $\exp(iS[x]/\hbar)$ where $S[x] = \int_{t_i}^{t_f} L(x, \dot{x}, t) dt$ is the classical action. The temporal interval $[t_i, t_f]$ over which the action is integrated is a finite interval, but the Lagrangian L is defined for all $t \in \mathbb{R}$ —it is a local function of the field and its time derivative at each moment, and its definition carries no restriction to positive times. More structurally, the amplitude $\langle x_f, t_f | x_i, t_i \rangle$ is itself a function of t_i and t_f ranging over $\mathbb{R}$; the case $t_i < 0, t_f > 0$ is as well-formed as any other, and the path integral over that interval is computed in exactly the same way. The path integral formalism imposes no half-axis restriction; the temporal endpoints are free parameters, and their range is all of $\mathbb{R}$.

In quantum cosmology, where the path integral is taken over geometries rather than particle trajectories, this point becomes sharper. The Hartle–Hawking no-boundary proposal (1983) formulates the quantum state of the universe as a path integral over compact Euclidean four-geometries, with the Euclidean time τ running over a compact interval determined by the geometry itself—there is no external restriction of τ to positive values, because the geometry determines the temporal structure rather than being defined against a pre-given temporal axis. Bounce cosmologies in loop quantum cosmology (Ashtekar, Pawłowski and Singh 2006) replace the classical singularity with a quantum bounce, and their effective equations have solutions extending continuously through $t = 0$ to $t < 0$. These remain theoretical proposals; the structural point they illustrate is that the path integral, when applied at the cosmological level, is not confined to the half-line by any feature of the formalism—the half-axis restriction would have to be imposed by hand, from outside the mathematics.

The cumulative picture across all six formalisms is unambiguous. The equations of classical and quantum dynamics, the Fourier apparatus used to analyse their solutions, the Green's functions and propagators that encode physical content, the time-reversal operator and its experimental consequences, the geometric structure of Minkowski spacetime, and the path integral in its quantum-cosmological

applications: none of these formalisms restricts the temporal variable to $[0, +\infty)$. The complete real axis is not a speculative extension of physics's mathematical apparatus; it is the apparatus. The interpretive restriction to the half-axis enters, without argument, only at the moment physics moves from calculation to explanation.

3. Three Justifications and a Category Error

If the formal domain is $(-\infty, +\infty)$, what justifies the interpretive restriction? Three answers recur in the literature and in the working attitudes of physicists. I take them in turn, examining each justification on its own terms and then identifying the counter-objection available within those terms.

3.1 The Thermodynamic Justification

The thermodynamic justification holds that the direction of entropy increase constitutes the physical reality of temporal direction, so that the "negative" region is simply the region physicists call the past, exhaustively described by forward-time physics run from earlier initial conditions. On this view, the half-axis restriction is not a restriction at all: $t < 0$ is available for use as "before $t = 0$," and the apparent truncation is merely a labelling choice about where the clock starts.

This response fails to engage the half-axis problem. The half-axis problem is not about labelling; it is about whether the temporal domain is the half-line $[0, +\infty)$ or the full line $(-\infty, +\infty)$. If $t < 0$ is fully available and means "before the reference event," then there is no half-axis restriction—and the question becomes why physics, in its interpretive statements, consistently treats the structure of time as one-ended rather than two-ended. If $t < 0$ is not available—if the physical universe has a first moment—then the justification for that unavailability is the cosmological one discussed in Section 3.3, not the thermodynamic one. The thermodynamic justification conflates the arrow of time with the extent of time.

The deeper failure, however, is structural. The second law is a statistical regularity of macroscopic systems whose microscopic laws are time-symmetric; the asymmetry is injected through a cosmological boundary condition—the Past Hypothesis (Albert 2000)—which is itself unexplained, and whose improbability under the standard measure Penrose (1989) estimated at one part in $10^{10^{123}}$. An asymmetry that is (i) statistical, (ii) macroscopic, and (iii) inherited from an unexplained boundary condition is an asymmetry within the domain. It has no resources to legislate about the domain.

The counter-objection available within thermodynamic terms makes this concrete. Suppose the Past Hypothesis is granted in full: the universe began in an extremely low-entropy state at $t = 0$, and entropy has increased monotonically since. This grants the thermodynamic justification everything it claims. It still does not follow that $t < 0$ is interpretively unavailable. The Past Hypothesis is a boundary condition

on a dynamical equation; boundary conditions do not determine the domain of the equation, they determine the solution within the domain. A cosmological model with a Past Hypothesis at $t = 0$ is fully consistent with a complete temporal axis on which $t < 0$ represents a pre-boundary region, whatever that region turns out to mean. To conclude from "entropy increases for $t > 0$ given the Past Hypothesis at $t = 0$ " that " $t < 0$ is physically empty" is to treat a boundary condition as a domain restriction—a confusion of two formally distinct concepts.

3.2 The Causal Justification

The causal justification identifies temporal direction with causal direction: the future is where effects are, the past is where causes are, and negative time is therefore just past time under another name. On this account, the half-axis problem dissolves into the arrow problem, and the arrow problem is to be solved by a theory of causal asymmetry.

The circularity of causal theories of temporal direction is well established (Price 1996, ch. 6; Mellor 1998, ch. 10): the asymmetry of causation is either primitive—in which case it is stipulated rather than explained—or grounded in temporal or entropic asymmetry, in which case the account presupposes what it purports to explain. But there is a further point specific to the half-axis problem that the standard circularity objection does not reach.

Even granting a non-circular causal arrow—granting, that is, that there is a fact of the matter about which direction in time is "toward causes" and which is "toward effects"—causal order is an ordering of events within the interpreted domain. An ordering relation on a set does not determine the extent of the set. The integers have a well-defined linear order, but the order does not entail that the set is bounded; the natural numbers have the same order type as the integers above any fixed element, but the natural numbers are bounded below and the integers are not, and no fact about the order relation within either set settles the question of the other's lower bound.

The causal justification therefore answers a question about structure within the temporal axis—which direction does causation run?—and is silent on the question actually posed—does the axis have a lower bound? These are not the same question. Establishing that causation runs from past to future does not establish that the past has a first moment, any more than establishing that the integers increase to the right establishes that there is a leftmost integer. The causal justification, like the thermodynamic one, addresses asymmetry and mistakes this for an answer about extent.

The counter-objection sharpens the point. Causal asymmetry and temporal asymmetry are related by a mutual dependence that the literature on both has not resolved (Price 1996; Woodward 2003; Frisch 2014). If causal direction is grounded in temporal direction, then causal asymmetry is a consequence of whatever asymmetry time possesses—and an asymmetry within the interpreted domain cannot generate, from within itself, a restriction on what the domain contains. If temporal direction is grounded in causal

direction, then the causal order is prior, and the temporal axis is whatever structure supports that order; but a causal order on an unbounded set supports the same local asymmetry as a causal order on a bounded set—the order is indifferent to the bound. In neither direction of the dependence does causal asymmetry constrain the extent of the temporal domain.

3.3 The Cosmological Justification

The cosmological justification is the most direct: time began at the Big Bang, so the physical domain of t is $[0, +\infty)$, with $t = 0$ the initial singularity; the negative half-axis is an artefact of the formalism with no physical referent. This justification, unlike the first two, does address the domain question rather than deflecting to the direction question. It therefore deserves correspondingly more detailed examination.

The Penrose–Hawking singularity theorems establish that, under the standard energy conditions, general-relativistic spacetimes of the cosmological class are geodesically incomplete toward the past (Hawking & Ellis, 1973)—that is, that past-directed timelike geodesics cannot be extended indefinitely but terminate at a boundary where the spacetime description becomes geodesically incomplete and the classical metric formulation ceases to provide an extension. This is a precise mathematical result. But it establishes that general relativity breaks down at that boundary, not that the boundary is a boundary of reality. The distinction is between a theory's domain of validity and the domain of the phenomena the theory describes. Geodesic incompleteness is a singularity of the theoretical description; the physical inference that follows is that a more complete theory is needed at the relevant scale, not that the phenomena stop.

This reading is not exotic. It is the explicit and consensus position of the quantum-gravity research programme. Bounce cosmologies in loop quantum cosmology (Ashtekar, Pawłowski and Singh 2006) resolve the classical singularity by quantum-geometric effects—the quantum bounce replaces the classical big bang, and the geometry extends to a contracting pre-bounce phase at $t < 0$ in the parametrisation natural to the model. Pre-Big-Bang scenarios in string cosmology (Gasperini and Veneziano 1993) extend the spacetime manifold through the string-scale regime to a pre-big-bang phase governed by the duality symmetries of the low-energy string effective action. Conformal cyclic cosmology (Penrose 2010) treats the classical singularity as a conformal boundary between aeons, with a well-defined geometric continuation. These proposals differ in their physical content and in their evidential status; they share the structural commitment that the classical singularity is a transition, not a terminus.

But the critical counter-objection does not require any of these models. It requires only the distinction between two propositions that the cosmological justification conflates:

(A) The Big Bang is a boundary condition of the classical relativistic cosmological model.

(B) The Big Bang is the ontological beginning of time.

These are not the same proposition, and no argument available within the resources of classical general relativity establishes (B) from (A). A boundary condition of a model is a constraint on the solutions of a differential equation within a specified domain; it says nothing about what exists outside that domain, because the model is not defined there. The inference from (A) to (B) is a category error of the same form as inferring, from the fact that Newtonian mechanics is undefined for velocities exceeding c , that velocities exceeding c do not exist. The failure of the theory at the boundary is evidence that the theory needs replacing; it is not evidence about what lies at or beyond the boundary, because the theory that would describe that region does not yet exist.

This point can be sharpened by attending to the Hawking–Penrose disagreement about the singularity, which the cosmological justification tends to ignore. Hawking (1988) proposed that the no-boundary condition in quantum cosmology—the Hartle–Hawking state—effectively removes the singularity by replacing the Lorentzian geometry near $t = 0$ with an Euclidean geometry in which "time" becomes a spatial dimension; on this account, the question "what came before the Big Bang?" is as malformed as "what is south of the South Pole?" Penrose (2010), by contrast, treats the classical singularity as a genuine boundary with a definite geometric continuation. The disagreement between two of the foremost contributors to singularity physics is not a settled matter; a foundational interpretive framework that rests its domain restriction on one side of this unresolved dispute does not stand on firm ground.

3.4 The Category Error and Its Common Structure

The three failures share a common structure, and identifying it is the central analytical contribution of this section. Each justification establishes, at best, a physical asymmetry: an asymmetry of entropic behaviour, of causal order, or of the domain of validity of a particular classical theory. Each then treats this physical asymmetry as though it effected a mathematical restriction: a contraction of the domain on which the temporal variable is defined and interpreted.

These are different kinds of claim. A physical asymmetry is defeasible—it can be superseded by a broader theory, a resolved singularity, a different boundary condition. A domain restriction, were it legitimate, would be a fact about the definition of the variable and thus indefeasible by physical developments. The justifications offer defeasible grounds and draw a conclusion that, if it is to be any conclusion at all, must be indefeasible. This mismatch—defeasible premises, indefeasible conclusion—is the category error.

Nothing in this analysis determines what the negative half-axis means or whether it has a physical interpretation. The analysis establishes that the three standard justifications for interpreting it away do not succeed, because none of them engages the right kind of question. A physical asymmetry within the

domain does not answer the question of what the domain is. Until an argument of the right kind is produced—one that addresses the domain directly, rather than deflecting to behaviour within it—the interpretive restriction is an unsupported convention.

4. The Inductive Argument: Negative Quantities and Regime-Limited Intuition

The half-axis restriction of time is not the first time physics has treated a quantity's mathematical range as truncated on the strength of regime-limited experience. The historical pattern is consistent enough to carry inductive weight, and it is worth developing beyond the three cases that are most frequently cited.

4.1 Temperature

Temperature was for two centuries a paradigmatically positive-definite quantity; the Kelvin scale seemed to make the positivity analytic, with absolute zero as an unreachable floor. Yet the statistical-mechanical definition of temperature, $1/T = \partial S/\partial E$, contains no positivity constraint: a system whose entropy decreases with energy—a population-inverted spin system—has negative absolute temperature (Ramsey, 1956). Purcell and Pound (1951) produced such a system in nuclear magnetic resonance; negative-temperature ensembles are now standard objects in quantum optics, and motional negative-temperature states have been realised in cold-atom lattices (Braun et al. 2013). The intuition of positivity was an artefact of the regime—unbounded energy spectra—in which thermometry happened to develop. Within that regime, no state with negative temperature was accessible; the impossibility was regime-bound, not mathematical.

4.2 Energy

Classical kinetic energy is positive-definite, and pre-relativistic physics treated the positivity of energy as constitutive. General relativity introduced negative gravitational binding energy as a standard concept; the total energy of a closed universe is plausibly zero by the argument of Tryon (1973), in which the positive energy of matter is exactly cancelled by negative gravitational energy—a proposal that has remained live in inflationary cosmology. Quantum field theory delivered locally negative energy densities: the Casimir configuration, in which vacuum fluctuations between two parallel conducting plates produce a negative energy density between them, was experimentally confirmed by Lamoreaux (1997). The positivity intuition encoded the regime of its formation—slow particles, negligible gravitational binding, no vacuum fluctuations—not a property of the quantity.

4.3 Spatial Position

Spatial position is the case that reveals the structure of the restriction most clearly. No physicist regards negative x , y , or z as problematic, because the arbitrariness of the spatial origin is transparent: the

coordinate's full range is used and interpreted without residue. A negative x-coordinate means "to the left of the origin I chose," and the physics is indifferent to the choice of origin. Space shows what a fully symmetric interpretive treatment of a coordinate looks like.

Time, formally identical in the Lorentzian signature to within a sign—the temporal coordinate entering the metric with opposite sign to the spatial coordinates—does not receive the same treatment. The formal parallelism is routinely invoked in relativity theory when it is convenient (Wick rotation, Euclidean field theory, conformal methods); it is abandoned, without argument, in the interpretive stance toward negative values. The asymmetry in treatment is not grounded in a formal difference between the coordinates; it reflects an asymmetry in the physical intuitions brought to the interpretation, intuitions that may well be regime-limited in the same way as those that initially restricted temperature and energy.

4.4 Negative Frequency

In classical wave theory, negative frequencies are formal artefacts: the condition that a real-valued signal $f(t)$ equals its own complex conjugate, $f(t) = f^*(t)$, entails that its Fourier transform satisfies $F(-\omega) = F^*(\omega)$, so that the negative-frequency components carry no independent information. Negative frequencies appeared to be a bookkeeping device, not a physical reality.

In quantum field theory, the status of negative frequencies is entirely different. The quantisation of the Klein–Gordon field decomposes the field into positive- and negative-frequency parts: $\phi(x) = \int d^3k/(2\pi)^3 [a(k) e^{-ikx} + a^\dagger(k) e^{ikx}]$, where the positive-frequency part is associated with particle annihilation operators $a(k)$ and the negative-frequency part with particle creation operators $a^\dagger(k)$. The physical distinction between particles and antiparticles, in the charged version of the field, is implemented through the assignment of positive and negative frequencies to different charge states. Negative frequencies are not eliminated by the quantisation procedure; they are promoted to physical significance as the Fourier components of the field that create particles from the vacuum. A formalism that regarded negative frequencies as physically meaningless would be unable to describe particle creation—that is, unable to describe the principal phenomenon of quantum field theory.

4.5 Dirac's Negative-Energy Solutions

The Dirac equation, $(i\gamma^\mu \partial_\mu - m)\psi = 0$, was derived in 1928 as the relativistic equation for the electron. Its solutions include, for every positive-energy solution with energy $E > 0$, a corresponding negative-energy solution with energy $E < 0$. In the context of 1928 physics, negative-energy electron states appeared physically impossible—an electron in a positive-energy state could radiate photons and cascade indefinitely downward through negative energies, violating conservation of energy. The solutions were mathematical objects without physical interpretation, and Dirac's initial response was to seek a way to eliminate them.

The resolution—the Dirac sea, and subsequently the concept of antiparticles—preserved the negative-energy solutions by giving them physical content. The filled Dirac sea interpretes the negative-energy states as occupied, and a "hole" in the sea corresponds to an antiparticle: the positron, predicted by Dirac (1931) and observed by Anderson (1932). In the modern quantum field theory reformulation, the negative-energy solutions are reinterpreted, in the Feynman–Stueckelberg picture, as positive-energy antiparticle solutions propagating backward in time—a reinterpretation that is itself a use of the negative temporal direction as a physically meaningful concept. The history of the Dirac equation is the history of a mathematical structure whose "unphysical" half—the negative-energy, or equivalently the negative-frequency, solutions—proved to encode the most unexpected physical content: the existence of antimatter.

4.6 Negative Curvature

Riemann's classification of constant-curvature geometries includes three cases: positive curvature (spherical), zero curvature (flat), and negative curvature (hyperbolic). In the context of nineteenth-century physics, positive curvature was familiar from the geometry of the sphere and plausible as a model of physical space; negative curvature was a mathematical curiosity whose physical relevance was unclear. In general relativity and modern cosmology, negative spatial curvature is a standard cosmological parameter: the curvature density parameter Ω_k can be positive, zero, or negative, corresponding to closed, flat, or open universes, and observational evidence—the CMB angular power spectrum—constrains Ω_k to be close to zero but is consistent with small negative values. Negative curvature is not a pathological extension of the geometric formalism; it is an empirically constrained physical possibility.

4.7 The Inductive Schema and Its Epistemic Weight

The six cases share a common pattern. A quantity is mathematically defined on a range that includes negative values. Experience within a limited regime—the regime in which the physics of the quantity was initially developed—gives no access to negative values. The mathematical structure is therefore either not noticed or actively interpreted as physically meaningless. Subsequent theoretical development, driven by the internal requirements of more comprehensive theories, reveals that the negative values are physically significant: they represent new states of matter (negative temperature), new physical effects (Casimir energy), necessary degrees of freedom (creation operators), or entirely new classes of entities (antiparticles). In each case the extension is made by the mathematics before it is made by the physics, and in each case resistance to the extension reflects the limits of the regime in which the quantity was initially understood.

The inductive schema is: (i) a quantity is mathematically defined on a range; (ii) experience within a limited regime suggests a truncated range; (iii) the truncation is reified into a quasi-definitional

restriction; (iv) the restriction is eventually recognised as regime-bound; (v) extension to the full range reveals new physics. Temperature, energy, spatial position, negative frequency, Dirac's negative-energy solutions, and negative curvature have completed this schema. Time stands at stage (iii).

The six cases do not, taken together, constitute a proof that negative time is physically meaningful. Inductive arguments of this form establish no such thing. What they establish is something more modest and more defensible: an epistemic heuristic. When a quantity is mathematically defined on a complete range, when the restricted region is inaccessible within the regime of the quantity's original development, and when the restriction has been reified without explicit argument, the default rational attitude is suspicion of the restriction rather than confidence in it. The heuristic is not a theory of induction; it is the distillation of a pattern whose recurrence across six independent cases of fundamental physics warrants treating as methodologically significant. To reject the heuristic in the case of time while accepting its deliverances for temperature, energy, and negative-energy solutions is to require a positive argument for the exemption—an argument that singles out the temporal axis as uniquely immune to the pattern. That argument has not been produced. The heuristic therefore applies, and the burden falls on the restriction.

5. The Structural Status of $t = 0$

Suppose the complete axis is taken interpretively seriously, as the formalism has always taken it computationally. One consequence follows immediately, and it concerns the point $t = 0$.

5.1 Three Characterisations of $t = 0$

On the half-axis reading, $t = 0$ is a boundary point of $[0, +\infty)$: topologically distinguished, but interpreted merely as the first moment—the initial condition, the singularity, the place where the account starts. On the complete-axis reading, $t = 0$ acquires a different and stronger characterisation, expressible in three equivalent ways.

Topologically, $t = 0$ is the unique point belonging to the closure of both open half-axes. Every neighbourhood of 0 intersects both $(-\infty, 0)$ and $(0, +\infty)$, and no other point of $\mathbb{R}$ has this property. It is the unique point of $\mathbb{R}$ that is a limit point of both the negative and positive rays simultaneously—the only point at which the two halves of the axis are in contact.

Algebraically, $t = 0$ is the identity of the additive group $(\mathbb{R}, +)$ under which the temporal axis has its natural structure. Every moment t has an additive inverse $-t$ satisfying $t + (-t) = 0$. The complete temporal axis is, in this precise algebraic sense, centred on 0: the point from which $+t$ and $-t$ are symmetric images of one another, and the only point invariant under the map $t \mapsto -t$ considered as a group automorphism. The half-axis $[0, +\infty)$ is not closed under additive inverses and does not form a

group; it lacks the algebraic structure that the complete axis possesses, and in particular it lacks an identity element in the additive group-theoretic sense—no point that stands in the same relation to all others that 0 stands to on the complete line.

Dynamically, $t = 0$ is the unique fixed point of the time-reversal operator $T: t \mapsto -t$. For any $t \neq 0$, $T(t) = -t \neq t$; only $t = 0$ satisfies $T(t) = t$. This is the one point of the temporal continuum that is invariant under the reflection whose symmetry properties physics measures with the experimental precision described in Section 2.4. The fixed point of a symmetry operation is not an arbitrary element of the domain; it is the element that the operation leaves unchanged, and in the context of a continuous symmetry group, fixed points characterise the structure of the symmetry in a way that no other points do.

5.2 Fixed Points and Physical Significance

The significance of fixed points of symmetry operations in physics is not confined to the temporal case. It is a general feature of the discipline's practice that the fixed points of fundamental symmetry operations are assigned determinate physical status, often serving as the natural reference states from which asymmetries are measured and against which departures require explanation.

The fixed point of spatial inversion $P: x \mapsto -x$ is the spatial origin, which in a translation-invariant theory is arbitrary but in a theory with a preferred centre—as in the hydrogen atom, where the proton defines a preferred origin—becomes physically distinguished. The charge-symmetric configuration—zero net charge, electrical neutrality—serves as the reference baseline from which the observed charge asymmetry of macroscopic matter is measured, playing the same structural role as a fixed point of the relevant symmetry operation. The fixed point of the combined CP operation in the space of matter-antimatter configurations is the symmetric state of equal matter and antimatter density, from which the observed baryon asymmetry of the universe—one of the principal unsolved problems of cosmology—represents a departure that demands explanation. In each case the symmetric fixed point is not merely a mathematical convenience; it is the structurally natural state, the state that the symmetry operation leaves unchanged, and therefore the state relative to which all asymmetric configurations are defined.

The fixed point of T on the temporal axis is $t = 0$. On the complete-axis reading, $t = 0$ is to the temporal structure what electrical neutrality is to the charge structure and what equal matter-antimatter density is to the baryon number structure: the point of maximal symmetry, the unique temporally direction-independent moment, the reference state from which all temporal asymmetries are departures. Physics has an established and systematic practice of asking, for each such fixed point, why the observed state departs from it—why there is net charge, why there is a matter surplus, why there is a preferred temporal direction. The half-axis reading of time makes the analogous question invisible, because a boundary point of a half-line is a terminus, not a centre of symmetry, and termini do not invite the question of why departures occur in both directions. Only on the complete-axis reading does the question become:

why is there a temporal asymmetry given that the temporal structure is centred on a point of maximal symmetry?

This is a more demanding version of the arrow-of-time problem than the standard one. The standard problem asks why there is a preferred temporal direction given that the microscopic laws are (approximately) time-symmetric. The complete-axis version asks why there is a preferred direction given that the temporal axis itself is centred on a direction-independent point. The two questions are related but not identical: the standard problem takes the half-axis domain as given and asks about asymmetric behaviour within it; the complete-axis version treats the symmetric point $t = 0$ as the natural reference and asks why the physical history is asymmetric about it. The complete-axis reading thus raises the explanatory standard, not by adding a new assumption, but by removing an unjustified one.

5.3 The Asymmetric Treatment of the Temporal and Spatial Origins

The contrast with spatial coordinates is instructive and largely unremarked. In spatial geometry, the origin $x = 0$ is treated as entirely conventional: any spatial point may serve as origin, the physics is invariant under spatial translation, and the negative half of each spatial coordinate axis is interpreted freely. The question "what is at negative x ?" is answered straightforwardly: whatever is to the left of the chosen origin.

The temporal origin $t = 0$ is not treated conventionally. It is identified with a specific physical event—the Big Bang, or the initial conditions of the system under study—and the negative region is either treated as inaccessible or ignored. This asymmetry in interpretive practice is not licensed by the formalism. The Lorentzian metric distinguishes the causal character of intervals—timelike, null, spacelike—but it does not distinguish the temporal origin from any spatial origin in a way that would justify interpretive asymmetry. In the Minkowski line element $ds^2 = -c^2dt^2 + dx^2 + dy^2 + dz^2$, the temporal coordinate and the spatial coordinates appear on fully parallel footing as coordinates of an event; the distinction is the sign in the metric, which affects causal structure, not the interpretation of the coordinate's range.

If the spatial origin's status as conventional is taken seriously—as it universally is—and if the formal parallelism between temporal and spatial coordinates in the Minkowski metric is taken seriously—as it is when computationally convenient—then the conclusion that the temporal origin is similarly conventional follows from combining two positions the discipline already holds. The asymmetry in interpretive practice is not a considered philosophical stance grounded in a formal disanalogy; it is an unconsidered habit whose continuation requires either a positive argument for the disanalogy or an acknowledgement that the habit is unjustified.

5.4 Structural Status Without Ontological Commitment

These are elementary mathematical facts—the topological, algebraic, and dynamical characterisations of $t = 0$ —and no ontological weight is claimed for them here beyond what they state. The argument of this section is not that $t = 0$ is a cosmologically significant moment, that it marks a phase transition in the laws of physics, or that the physical history is symmetric about it. The argument is prior and narrower: that the complete-axis reading assigns $t = 0$ a determinate structural status for which the interpretive framework of physics currently has no account and recognises no obligation to provide one.

On the half-axis reading, $t = 0$ is a terminus, and no structural account of a terminus is required: a terminus is simply where the domain starts, and the question of why it starts there is deflected to the cosmological or thermodynamic justifications examined in Section 3. On the complete-axis reading, $t = 0$ is a centre of symmetry, and centres of symmetry in physics require accounts: they are the points from which departures demand explanation, and the discipline has a systematic practice of providing such explanations. The half-axis convention has not merely truncated a range; it has suppressed a question. Any complete interpretation of the temporal formalism owes an account of the fixed point of the time-reversal operator. The current interpretive framework does not have one, and has not noticed that one is owed.

6. Relation to Existing Programmes, Scope, and Directions

6.1 The Archimedean Programme and the Present Extension

The argument closest in spirit to the present one is Price's (1996) Archimedean programme: the demand that the analysis of temporal asymmetry be conducted from an "atemporal standpoint" not itself infected by the asymmetries under analysis. The present paper is aligned with Price's methodological demand and can be read as extending it one level down. Price applies the atemporal standpoint to the asymmetries of physical process—radiation, thermodynamics, causation—and exposes double standards in their standard treatments: physicists use different inferential standards when reasoning toward the past and toward the future, without justification, and the Archimedean standpoint requires that the same standards be applied in both directions.

The half-axis problem applies the same standpoint to the constitution of the temporal domain itself, and exposes a double standard more basic than any Price treats: the discipline computes on a symmetric domain and interprets an asymmetric one. Price's target is the double standard in the treatment of directions within the domain; the half-axis problem's target is the double standard in the treatment of the domain's extent—the use of $(-\infty, +\infty)$ in computation and $[0, +\infty)$ in interpretation. The arrow literature, Price included, has been a debate about the direction of time. The half-axis problem concerns

the extent of time. Extent is prior to direction, because a direction is a structure on a domain already given.

The relationship is not one of conflict but of complementarity. A satisfactory response to the half-axis problem—an argument that entitles the interpretive restriction to the half-axis, or an argument for the full axis—would change the domain against which Price's analysis is conducted. If the complete axis is warranted, then the Archimedean demand applies not only to the asymmetries of processes at positive times but to the relationship between the positive and negative temporal regions. The two problems are nested, with the half-axis problem at the outer level.

6.2 What Has Been Established

It is worth being precise about what the foregoing has and has not established. It has not established that negative time is physically real, that the universe extends through $t < 0$, or that any particular cosmological model is correct. The argument is diagnostic and destructive, not constructive; it establishes that the current interpretive framework has not paid for its restriction, not what the correct alternative is.

Specifically, the paper has established three things.

First, the interpretive restriction of time to the half-axis is supported by none of the justifications standardly assumed to support it. The thermodynamic justification conflates asymmetry within the domain with restriction of the domain, and the counter-objection shows that the Past Hypothesis, even if granted, is a boundary condition that does not settle the extent of the domain. The causal justification confuses an ordering relation on a set with a constraint on the set's extent, and the counter-objection shows that the mutual dependence of causal and temporal asymmetry is indifferent to the question of a temporal bound. The cosmological justification conflates the boundary of a theory's domain of validity with the boundary of reality, and the counter-objection—internal to the physics of singularities—shows that the singularity theorems establish geodesic incompleteness of the relativistic description, not the ontological terminus of time.

Second, the history of physical quantities gives inductive grounds for suspecting regime-limited truncations of formally complete ranges, and time exhibits every diagnostic feature of the pattern: mathematical completeness, regime-limited inaccessibility of the restricted region, reification of a convention into a quasi-definitional restriction. Temperature, energy, spatial coordinates, negative frequency, Dirac's negative-energy solutions, and negative curvature all completed the pattern by extension. The onus of argument falls on the position that time is uniquely exempt.

Third, the complete-axis reading assigns the point $t = 0$ a determinate structural status—fixed point of time reversal, additive identity, topological centre of the complete line—for which the interpretive framework of physics currently has no account and recognises no obligation to provide one. The half-

axis convention has not merely truncated a range; it has rendered invisible a structure that the mathematics defines precisely and that physics's own practice—in symmetry analysis, in the assignment of physical significance to symmetric zeros—is committed to taking seriously.

6.3 Directions for Further Investigation

Three questions arise naturally from the analysis and are left for further work.

The first concerns quantum cosmology. If the complete temporal axis is taken interpretively seriously, the boundary conditions of quantum cosmological models—including the Hartle–Hawking no-boundary proposal and the tunnelling proposal of Vilenkin (1982)—require re-examination. These proposals are formulated as conditions on the path integral over geometries; they do not, in their standard formulations, take a position on the extent of the temporal axis prior to the quantum-cosmological calculation, because that extent is determined by the calculation. The question of whether consistent quantum-cosmological boundary conditions can be formulated on a complete axis, and what their physical implications would be, is open.

The second concerns the interpretation of $t = 0$. The structural analysis of Section 5 assigns $t = 0$ a determinate formal status—fixed point of time reversal, additive identity, topological centre—but does not assign it a physical interpretation. The range of physically distinct interpretations available within the complete-axis framework is not obvious. The point might be an unobservable structural symmetry, an observational horizon, a phase transition in the properties of physical law, or the locus of a cosmological symmetry whose breaking generates the observed temporal asymmetry. Distinguishing these possibilities requires a theoretical framework that takes the complete-axis reading as a starting point.

The third concerns the observational status of negative time. The diagnostic features of the inductive pattern—new physics revealed by the extension—suggest asking whether negative time has observational consequences distinguishable from those of a pure half-axis cosmology. This is a question for physical modelling and does not admit of a general answer at the foundational level. It is nonetheless the question that a positive extension of the present negative argument would eventually face, and framing it precisely is part of the productive agenda opened by clarifying the half-axis problem.

6.4 The Debt

The conclusion is deliberately austere. Physics is entitled to its computational use of the complete axis; it has earned that entitlement in three centuries of empirical success. It is not entitled to the interpretive deletion of half of it, because it has never paid for that deletion in argument. The debt has been invisible because the convention is old. It is nonetheless a debt. The half-axis problem is the demand that it be either paid or acknowledged as unpayable—and either outcome would constitute progress in the

foundations of the subject, because a discipline that knows what it has assumed is in a different epistemic position from one that has forgotten that an assumption was made.

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